

The State of Performance Art and Interactive Technology

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Abstract

The workshop proposed here will focus on performance art or installation art that incorporates aspects of wearable technology, sensors, or incorporation of AI. We hope to facilitate presentations and discussions between artists, designers, engineers, and technologists with the goal of sharing innovative interactions and uses of wearable technology in creative ways. This will be an open workshop but will include a call for papers for those that want to publish concepts, methods, or project examples of state-of-the-art use of technology in interactive performance or installation art pieces.

CCS Concepts

• **Human-centered computing** → *Interaction devices*;

Keywords

Performance Art, Dance, Choreography, Art, Design, Wearable Artistic Interactions

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1 Motivation

When asked “Why work on performance led research in the wild” Benford et. al give three reasons. First among them is that Artists’ uses of emerging technologies are often highly innovative and unusual, stretching the technology in unforeseen ways, highlighting new design values and approaches that are sometimes contrary to

received wisdom in HCI (e.g., ambiguity or discomfort), and opening up new areas of application. [2]

Performance artists working with wearable technology experts has a fruitful and rich history. [14, 18, 21] Musicians, dancers, choreographers, drag queens, magicians, installation artists, theater actors and productions among others could all benefit from technological innovations to create more immersive and interactive art. Computing researchers working with artists benefit from stepping outside their typical methodologies. Both technology researchers and artists broaden their vernacular and empathy toward other disciplines when they work together on interesting and engaging projects.

Aside from wearable sensor interactions, many new performance and audience interactions are being created using computer vision motion capture and AI. [4] These new techniques and interactions are worth documenting and archiving for the ISWC and UbiComp community. It is also important to have venues for artists and performers to engage with the computing and technical academic community to build relationships for future collaborations.

2 Workshop

2.1 Workshop Format and Intended Audience

The workshop will focus on presenting new and innovative uses of technology in art (interactive performance and installation). We hope to accommodate up to 30 people (although we could accommodate more if interest is great and we have a large number of submissions). Anyone interested in art, performance art (which could include, but are not limited to, dance, music, and interactive installations), wearable technology, or teaching courses to students on wearable technology is encouraged to attend. No special skills are required. This will be an *open workshop* under a lightweight juried model with a short review schedule. We hope to have some papers to include in the adjunct proceedings that might take the form of project descriptions, case studies, and preliminary research findings. We also invite participants to bring demos and artifacts for a show and tell. This gadget show could include full-blown prototypes or examples of novel uses of off the shelf technology.



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Figure 1: Le Monstre interactive performance garment used a number of touch and movement sensors to interact with audience members. Photo by Safety Third Productions [19]

We are excited to bring together academics and professionals from both the arts and wearable technology communities.

2.2 Impact, Benefits and Learning Goals

We expect this workshop to be an opportunity for artist and designers along with computing researchers to showcase projects, outcomes, and their experiences creating new and innovative technology-enabled interactions for performance art. Although there will be a call for papers, we hope to facilitate non-paper presentations through video and discussion. The hope is that this workshop will open the ISWC and UBiComp community to artists in an inclusive and inviting way. Artists wishing to attend do not have to be already well versed in wearable technology or interdisciplinary work; in fact, all that is required is an interest in creating interactive technology.

We see this workshop as an opportunity to learn new techniques, methodologies, and experiences from each other. We hope to build

a list of new interesting devices that are on the market and catalog novel uses and online communities of makers and innovators willing to give advice. A main outcome of the workshop should be networking and building possible collaborations between attendees, nurturing the ISWC design community, and perhaps bolstering future submissions to the ISWC Design Exhibition.

2.3 Workshop Schedule Half-day

This will be a half-day workshop unless interest is great and there is space to accommodate a larger number of participants.

Overview and Presentation – 30 min

State of Performance Art and Technology Paper Presentations – 60-90 min

If we have a large number of submissions we could move to a full-day workshop.

Show and Tell - Costumes, Devices, Off the shelf hacks, methods and techniques. We want to leave room and time for both those who presented papers and those who just brought prototypes and demos to present and share. **60 - 90 min**

Discussion and Networking – 60 - 90 min

Discussion of Possible White Paper on the State of Performance Art and Wearable Technology including a collection of new academic references through a literary review. - **30 min**

2.4 Pre-Workshop Activities

We have a rather large number of workshop organizers in order to be able to advertise our workshop to different communities in different regions. We will meet virtually monthly before the workshop to encourage submissions and attendance from both artists and technical participants. Seeing as though we have such a large number of organizers, we will also run the lightweight review mostly internally without the need for a large recruitment of external reviewers.

3 Organizer's Backgrounds

Clint Zeagler

Zeagler is a Principal Research Scientist and Director of Strategic Partnerships for the Institute of People and Technology IPaT at Georgia Tech. Zeagler has been an active member of the ISWC community for years helping organize the ISWC design exhibition and leading a workshop at ISWC 2015 entitled "Wear and Tear: Constructing Wearable Technology for the Real World". He has also lead and organized workshops and tutorials at other conferences. [11, 17, 20?] Zeagler has published a number of papers on the topic of technologists working with performance artists and has a network of both artists and academics. [18, 19, 21]

Lucy Dunne

Dunne is a Professor and the Apparel Design Program Director of Apparel Design and Apparel Studies Track of Design Graduate

Program Wearable Technology Lab Co-Director at the University of Minnesota. She has also been instrumental in the support and creation of ISWC Design Exhibition. Dunne's expertise in fashion and apparel design make her a great addition to the workshop team. [3, 16]

Maribeth Gandy Coleman

Gandy is a Regent's Researcher, Assistant Vice Provost for Research Faculty, and the Director of Research for IPaT. In her 20+ as a research faculty member her work has been focused on the intersection of technology for mobile/wearable computing, augmented reality, human computer interaction, assistive technology, and gaming. Her work includes working with groups like Duran Duran in augmented reality experiences for audiences through image projection. [5, 18]

Katherine Hellen Fisher

Fisher is a dancer, choreographer, curator, creative producer, and Emmy Award-nominated film director. Her work interrogates the boundaries between digital and physical worlds through performance and technology. She creates participatory installations that center agency, empowerment, and desire while imagining radical, techno-feminist futures. [19] <https://www.katherinehellenfisher.com>

Corian Ellisor

Corian Ellisor is an Atlanta based performance artist with a concentration on dance theater. Professionally, Ellisor has worked with a variety of arts communities locally and abroad: Georgia, Texas, Florida, Massachusetts, Washington DC, New York, Guatemala, Sweden, The Netherlands, Germany and The United Kingdom. He uses his art as activism to create abstract narratives using dance, theater, drag, music, costumes and storytelling. Ellisor has generated over 50 performance works throughout his career including eleven evening length productions. Ellisor was selected as the Institute for People and Technology's (IPaT) 2025 artist-in-residence. Ellisor will engage with IPaT academics through a structured program to translate, highlight, incorporate, and interpret ongoing research through artistic endeavors utilizing the IPaT Craft Lab. <https://research.gatech.edu/corian-ellisor-selected-ipat-artist-residence>

Raja Schaar

Schaar, IDSA is Associate Program Director and Assistant Professor of Product Design at Drexel University's Antoinette Westphal College of Media Arts and Design. Raja's interdisciplinary research focuses on addressing inequities in maternal health; methods for engaging black girls and underrepresented minorities in STEM/STEAM through design and technology; innovation and entrepreneurship education; and biologically-inspired design and sustainability. [1, 12, 13]

Noah Posner

Noah Posner is a research scientist with the Institute for People and Technology and lab manager of the Interactive Product Design Lab (IPDL) in the School of Industrial Design at Georgia Tech. Posner's

focus is on creating interactive experiences that leverage physical interaction. He holds degrees in Industrial Design, Mechanical Engineering and an M.S. in Human Computer Interaction. His research involves creating rich physical interactions, designing and fabricating interactive devices using CAD and Rapid prototyping technologies, and developing educational tools for STEAM learning that focus on utilizing hands on learning with intentionally designed artifacts. He also teaches courses in Physical Prototyping for Human Computer Interaction and Industrial Design. [6, 10]

Jamie Ward

Jamie is a Senior Lecturer at Goldsmiths where (since May 2024) he is acting Head of Department for Computing. He sits as a staff representative on the college Academic Board, and as a board representative to the governing Council. Between academic positions Jamie worked variously as a technology consultant, an analogue circuit designer, and as an actor in television, film, and theater. [7, 8, 15]

Lisa Marks

Lisa Marks is an Associate Professor in the School of Industrial Design at Georgia Tech. She teaches studio courses in both undergraduate and graduate programs. Marks' research focuses on combining endangered and traditional handcraft with algorithmic modeling to create new production methods. [9] <https://lisamarks.com>

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